

EVALUATION OF A PHARMACEUTICAL CARE MODEL ON DIABETES MANAGEMENT

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OBJECTIVE: To assess the effectiveness of a pharmaceutical care model on the management of non-insulin-dependent diabetes mellitus (NIDDM) in urban African-American patients.

DESIGN: Eligible patients were randomized to either a pharmacist intervention or control group and followed over a 4-month period. Patients in the intervention group received diabetes education, medication counseling, instructions on dietary regulation, exercise, and home blood glucose monitoring, and evaluation and adjustment of their hypoglycemic regimen. Patients in the control group continued to receive standard medical care provided by their physicians.

SETTING: A university-affiliated internal medicine outpatient clinic.

PARTICIPANTS: The study population consisted of urban African-American patients with NIDDM currently attending the clinic.

MAIN OUTCOME MEASURES: Primary outcome measures included fasting plasma glucose and glycated hemoglobin concentrations. Secondary outcome endpoints included blood pressure, serum creatinine, creatinine clearance, microalbumin to creatinine ratio, total cholesterol, triglycerides, high-density lipoprotein, and low-density lipoprotein concentrations. Quality-of-life assessments were performed in both groups at baseline and at the end of the study.

RESULTS: Thirty-nine patients (17 intervention, 22 control) completed the study. The intervention group consisted of 12 women and 5 men with a mean $\pm$ SD age of 59 ± 12 years, total body weight (TBW) of 93 ± 22 kg, body mass index (BMI) of 34 ± 7 kg/m², and duration of NIDDM 6.8 ± 6.5 years. The control group consisted of 15 women and 7 men with a mean age of 65 ± 12 years, TBW of 88 ± 19 kg, BMI of 33 ± 7 kg/m², and a duration of NIDDM of 6.2 ± 4.8 y. Significant improvement in glycated hemoglobin ($p = 0.003$) and fasting plasma glucose ($p = 0.015$) was

achieved in the intervention group. No change in glycemia was observed in the control subjects. Statistically significant differences in the final glycated hemoglobin ($p = 0.003$) and fasting plasma glucose ($p = 0.022$) concentrations were noted between groups. No significant changes in blood pressure control, lipid profile, renal function parameters, weight, or quality-of-life measures were noted within or between groups.

CONCLUSIONS: Our data demonstrate the effectiveness of pharmaceutical care in the reduction of hyperglycemia associated with NIDDM in a group of urban African-American patients.

Ann Pharmacother 1996;30:238-43.

THE PRESENT TREND in health care to shift the focus from inpatient to ambulatory care and from disease treatment to disease prevention has created new opportunities for pharmacists to provide complete pharmaceutical services in ambulatory care settings. Indeed, new capitative and prospective methods of reimbursement have prompted hospitals to reduce length of stay in hospitals and, whenever possible, to avoid hospitalizations. Hospital inpatient days have decreased by 18.3% and outpatient visits increased by 34.4% from 1979 to 1989 in the US.¹ A number of pharmacists have seized the opportunities created by the growth of the ambulatory primary care. In a survey published in 1993, the extent to which hospital-based pharmacists provided ambulatory clinical services in the US was explored.² Of the 1174 pharmacy directors who returned the survey, 19% stated that pharmacists provided patient care in one or more ambulatory clinics at their institution. Despite the evolution of direct patient care services provided by pharmacists, the provision of pharmaceutical care in ambulatory settings is still believed to be incomplete.³ Studies documenting the positive effect of ambulatory care pharmacy services on patient outcomes and healthcare costs support efforts to extend pharmacist-provided clinical activities.⁴⁻⁸

Diabetes mellitus has been estimated to affect 6.6% of the US population.⁹ The prevalence of diabetes is far greater among African-Americans, Mexican-Americans, and Indian-Americans compared with whites.^{9,10} The incidence of diabetes is also rising more rapidly in these minority groups than in whites.¹⁰ Published pharmaceutical care models in diabetes management have been primarily

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This work was supported by a grant from the Diabetes Research and Education Foundation and Upjohn. The data were presented at the Winter Practice and Research Forum of the American College of Clinical Pharmacy, February 14, 1995, Orlando, Florida.

limited to patient education, compliance enhancement, and therapeutic drug monitoring activities.¹¹⁻¹⁵ The objective of the current study was to determine the impact of a comprehensive pharmaceutical care model on the treatment outcome measures of non-insulin-dependent diabetes mellitus (NIDDM) in urban African-American patients.

Methods

SUBJECTS

Urban African-American patients with NIDDM currently attending a university-affiliated general internal medicine outpatient clinic were considered for inclusion after medical chart review. The criteria for exclusion included insulin-dependent diabetes mellitus, renal dysfunction (serum creatinine >133 $\mu\text{mol/L}$), hepatic disorder (concentration of serum aminotransferases 3 times above normal), significant cardiac complications within the last 6 months, mental incompetence, or a history of noncompliance with regular clinic visits within the last 2 years. Subjects who met the study criteria were contacted by a letter which was followed up by a telephone call. Written informed consent was obtained from all subjects before admission to the study. A complete medical history, physical examination, and screening biochemical tests were obtained at that time.

DESIGN

Eligible patients were assigned to an intervention or control group in a randomized, parallel design fashion and followed over a 4-month period. For the intervention group, all diabetes-related management aspects were solely provided by a pharmacist. Patients in the intervention group received pharmaceutical care consistent with but broader than that described by Helper and Strand.¹⁶ This disease-specific model of pharmaceutical care included: (1) diabetes-specific pharmacotherapeutic evaluation and dosage adjustments, (2) comprehensive and individualized patient education regarding diabetes and its complications, (3) training on the recognition and treatment of hypoglycemia and hyperglycemia, (4) medication counseling, (5) specific instructions on dietary regulation and an exercise plan, and (6) training for self-monitoring of blood glucose (SMBG). These patients were individually taught SMBG and schedules detailing written instructions for specific testing times relative to meal consumption were provided. Patients were further instructed to record the date and time of any hypoglycemic events as well as symptoms they experienced during these events. Evaluation of current hypoglycemic therapies were performed during each clinic visit based on the self-tested blood glucose responses performed 4 times per day for 2 days per week. Hypoglycemic therapeutic regimens were adjusted or titrated to achieve fasting blood glucose (FBG) concentrations of ≤ 6.6 mmol/L (≤ 120 mg/dL) and 2-hour postprandial blood glucose concentrations of <10.0 mmol/L (<180 mg/dL) or to reach maximum daily doses of the sulfonylurea agent. Follow-up was conducted on a scheduled weekly basis until targeted glycemic control was reached. Thereafter, patients came to the clinic every 2–4 weeks for the duration of the study. On each clinic visit, patients were questioned in a nonthreatening manner regarding medication usage and asked to show their SMBG records. These measures were consistently taken to determine and ensure medication adherence, patient understanding, and self-engagement in diabetes control.

Patients in the control group reported to the clinic on two occasions: once for the initial assessment and once for the final exit visit. They were instructed to continue to receive standard care from their primary care providers. The customary medical care standards of this clinic involved seeing patients every 3–4 months to provide care as needed.

All physical examinations were performed by two medical residents. A pharmacist in charge was delegated full prescribing authority under a Human Investigational Committee–approved written protocol regarding hypoglycemic agents. Clinical, laboratory, and quality-of-life assessments were obtained from all subjects in both groups at baseline and at the end of the study. Primary therapeutic outcome measures included fasting plasma glucose (FPG) and glycated hemoglobin (GHb) concentrations. Secondary outcome endpoints included blood pressure, serum creatinine, creatinine clearance, microalbumin to creatinine ratio, total

cholesterol, triglycerides, high-density lipoprotein, and low-density lipoprotein concentrations. FBG, weight, and blood pressure measurements, along with review of home blood glucose monitoring values, hypo- and hyperglycemic episodes evidenced by symptomatology and/or SMBG values, and patient compliance with medications, diet, and exercise plans were obtained at each clinic visit.

LABORATORY AND ENDPOINT MEASURES ANALYSIS

Home blood glucose monitoring was performed with the One Touch Basic glucose reflectance meter (Lifescan, Milpitas, CA). Urine for microalbuminuria and creatinine clearance measurements was collected over a 12-hour period starting the evening before the scheduled appointment. Microalbuminuria was defined as urinary albumin of 20–200 $\mu\text{g/mg}$ creatinine.¹⁷ Glucose concentrations were measured with the Ektachem calorimetric test (Eastman Kodak, Rochester, NY). Glycated hemoglobin concentrations were determined with the Isolab Glyc-Affin test kit (Isolab, Akron, OH); the normal range was reported as 4.0 – 8.0% (mean $\pm$ SD $6\% \pm 1\%$).

Health-related quality of life (HRQOL) was assessed with the Health Status Questionnaire version 2.0, a validated general health assessment instrument derived from the Short Form (SF)-36 with documented reliability and scoring technique.¹⁸ Patients were instructed to complete the survey without assistance. Few patients required the survey to be read to them. At no time were the patients offered any interpretation of questions. The general survey was chosen to assess the impact of diabetes and its management on the patients' sense of well-being. SF-36 surveys have been previously used and validated in the NIDDM population.^{19,20} SF-36 measures physical and social functioning, role limitations, mental health, energy level, and general perceived health status.

STATISTICAL ANALYSIS

Statistical analysis was performed by using the computer software SPSS 6.0 for Windows (SPSS, Chicago, IL). Statistical differences in outcome measures between the two groups were tested using Student's *t*-test for independent samples. Differences in baseline and endpoint outcome measures were determined by using the paired *t*-test statistical analysis. Categorical data were tested by using chi-square analysis. Quality-of-life scores were analyzed with the same statistical tests. All data are expressed as mean $\pm$ SD. A *p* value of less than 0.05 was considered statistically significant.

Results

In review of 891 medical charts, 156 patients were identified who met the inclusion criteria of the study. Of those, 84 were not interested and 27 were not reachable. The remaining 45 patients were recruited and randomized to either the intervention or control group. Six patients in the intervention group dropped out or were discharged from the study. Of those, 4 found it difficult to comply with the frequency of clinic visits, 1 was discharged by the study investigators because of unstable angina within the first 2 weeks of the study, and 1 was lost to follow-up. Thirty-nine patients completed the study.

The intervention group consisted of 17 patients (12 women, 5 men) with a mean age of 59 ± 12 years and GHb of $11.5\% \pm 2.9\%$. The control group consisted of 22 patients (15 women, 7 men) with a mean age of 65 ± 12 years and GHb of $12.2\% \pm 3.5\%$. Hypertension was present in 14 patients (82%) in the intervention group and 17 (77%) in the control group. Obesity, defined as a body mass index of more than 27 kg/m^2 , was noted in 13 patients (76%) in the intervention group and 18 (82%) in the control group. Lipid abnormalities were observed in 10 patients (59%) in the intervention group and 11 (50%) in the control group. Patients received 2.8 ± 1.7 and 2.7 ± 1.3

medications in the intervention and control groups, respectively. Pertinent patient demographic characteristics are depicted in Table 1. The two groups were essentially similar at baseline.

In the intervention group, 38 pharmacotherapeutic interventions were made in oral hypoglycemic regimens (mean of 2.2 interventions per patient), including 25 dosage increases, 6 dosage reductions, and 7 alterations in medication. At the onset of the study, 10 patients were receiving maintenance doses of glipizide (Glucotrol), 5 glyburide (Micronase), 1 glyburide (Diabeta), and 1 chlorpropamide (Diabinese). During the course of the study, the drug therapy was switched from Glucotrol to Glynase in 3 patients, Glucotrol to Micronase in 2 patients, and Micronase to Glynase in 2 patients because of inadequate control. In comparison, 9 pharmacotherapeutic interventions (mean of 0.4 interventions per patient) were reported in the control group. These included 4 dosage increases, 1 dosage reduction, and 4 alterations in medication. The mean number of visits was 9.1 ± 2.7 in the intervention group and 2 in the control group.

Glycemic control variables are shown in Table 2. Baseline and final GHb concentrations for each subject are illustrated in Figure 1. Glycemic control was significantly improved in the intervention group and remained unchanged in the control subjects. A significant decline in final FPG (8.5 ± 2.3 vs. 11.1 ± 4.0 mmol/L, $p = 0.015$) and GHb ($9.2\% \pm 2.1\%$ vs. $11.5\% \pm 2.9\%$, $p = 0.003$) concentrations compared with baseline was observed in the intervention group. A statistically significant difference in the final FPG ($p = 0.022$) and GHb ($p = 0.003$) responses as

well as the mean absolute change in GHb ($p = 0.021$) between the two groups was reached at the end of the study. Figure 2 depicts the baseline GHb versus the change in GHb from baseline to the final measurement for each subject in the intervention group. The magnitude of change correlated significantly with baseline GHb measurements ($r = 0.72$, $p = 0.0011$). The graph further suggests that a greater reduction in final GHb occurred in subjects with higher baseline GHb concentrations.

Blood pressure measurements did not differ significantly between or within groups. However, a trend toward a statistically significant ($p = 0.07$) decline in the final systolic (140 ± 20 vs. 147 ± 23 mm Hg) and diastolic (82 ± 10 vs. 88 ± 9 mm Hg) blood pressure values compared with baseline was observed in the intervention group. No significant changes in body weight, serum lipid measurements, and renal function parameters were noted between or within groups with the exception of a decrease in the final serum creatinine (88 ± 18 vs. 97 ± 18 μ mol/L, $p = 0.005$) and a tendency toward an increase in the final low-density lipoprotein (3.54 ± 0.28 vs. 3.23 ± 1.32 mmol/L, $p = 0.023$) concentrations compared with baseline in the control group. All of these secondary endpoints values were within normal ranges, with a less than 2% change occurring during the study.

Quality-of-life analyses revealed no significant differences in any of the domains tested between or within groups. Reliability analysis indicated a high degree of internal consistency for all domains, ranging from 0.66 to 0.87 (Cronbach's alpha).

There were 17 reported hypoglycemic reactions in the intervention group and 2 in the control group. All the reactions were considered mild to moderate in nature; they

Table 1. Baseline Patient Characteristics*

CHARACTERISTIC	CONTROL GROUP	INTERVENTION GROUP
Number	22	17
Age (y)	65 ± 12	59 ± 12
Sex (female/male)	15/7	12/5
Total body weight (kg)	88 ± 19	93 ± 22
Body mass index (kg/m ²)	33 ± 7	34 ± 7
Duration of diabetes (y)	6.2 ± 4.8	6.8 ± 6.5
Fasting plasma glucose (mmol/L)	12.7 ± 4.7	11.1 ± 4.0
Glycated hemoglobin (%)	12.2 ± 3.5	11.5 ± 2.9
Serum creatinine (μ mol/L)	97 ± 18	106 ± 44
Mircalalbumin:creatinine ratio (μ g/mg Cr)	19.3 ± 24.4	11.4 ± 13.7

Cr = creatinine.

*Data given as mean $\pm$ SD. None of the characteristics were significantly different at baseline.

Table 2. Glycemic Control Parameters at Baseline and End of Study

VARIABLE	CONTROL GROUP	INTERVENTION GROUP
Fasting plasma glucose (mmol/L)		
baseline	12.7 ± 4.7	11.1 ± 4.0
final	11.0 ± 3.9	$8.5 \pm 2.3^{a,b}$
change	1.8 ± 3.9	2.6 ± 3.4
Glycated hemoglobin (%)		
baseline	12.2 ± 3.5	11.5 ± 2.9
final	12.1 ± 3.7	$9.2 \pm 2.1^{a,b}$
change	0.1 ± 3.0	2.2 ± 2.6^b

^aSignificantly different within a group ($p < 0.05$).

^bSignificantly different between groups ($p < 0.05$).

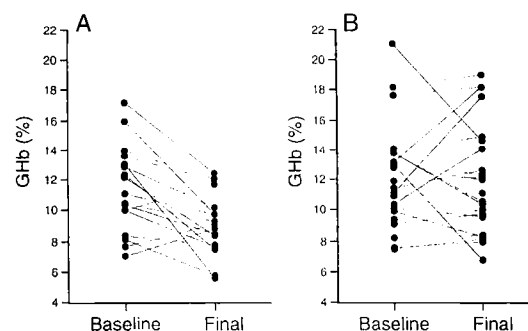


Figure 1. Baseline and final glycated hemoglobin (GHb) concentrations in the intervention (A) and control (B) groups.

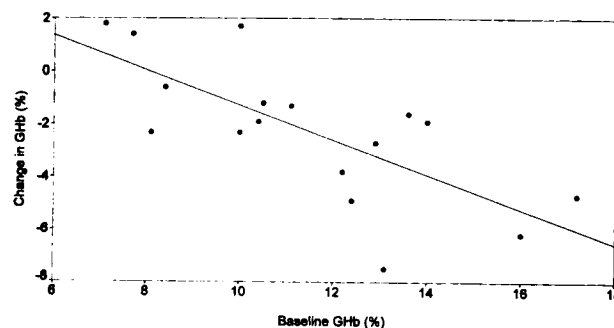


Figure 2. Linear regression of baseline versus change in glycated hemoglobin (GHb) concentrations in the intervention group.

were manifested by classic autonomic symptoms, recognized by patients, and successfully self-treated.

Three patients were hospitalized during the course of the study. Two patients in the control group were hospitalized, one for elective surgery and one for an unspecified leg problem, and one in the intervention group for an episode of chest pain.

Discussion

The provision of pharmaceutical care is a concept adaptable to patient-specific needs and the care setting. The future of pharmaceutical care in an ambulatory care setting is evolving. Documentation of the effectiveness of such pharmacist-provided direct patient healthcare services in improving clinical outcome, preventing complications, and reducing costs is fundamental for the expansion and maturation of pharmaceutical care. Our model provides effective individualized direct patient care services in an ambulatory care setting.

This study was designed to examine the effectiveness of a pharmacist-run clinic model on the degree of glycemic control in a group of urban African-American patients with NIDDM, a population at increased risk for diabetes and its complications along with identifiable barriers to adequate health care. Glycemic control as evidenced by FPG and GHb measurements significantly improved in those patients receiving pharmaceutical care interventions. Patients with poorest control experienced the greatest reduction in hyperglycemia. Given that the observed improvement in glycemia occurred in the absence of any significant changes in body weight, the impact of dietary or exercise instructions appears to have been minimal. Optimizing the oral hypoglycemic therapeutic regimens and enhancing patient understanding of the disease were most likely responsible for the observed reduction in hyperglycemia. Secondary outcome measures such as lipid abnormalities, body weight, blood pressure control, and renal disorders were not significantly changed during the course of the study.

Of particular importance in the current study is the magnitude of reduction of glycemia. The landmark Diabetes Control and Complication Trial²¹ (DCCT) firmly established that intensive glycemic control results in a dramatic and significant decrease in the rate and progression of microvascular complications and clinical neuropathy associated with insulin-dependent diabetes mellitus. The extent to which the results of the DCCT can be extrapolated to patients with NIDDM in terms of achieving normoglycemia remains a subject of debate. The DCCT established a direct relationship between the degree of glycemia and the development of diabetes-related complications so that a reduction in baseline glycemia will have beneficial effects.²² The applicability of the DCCT management approach to the general public is quite difficult. Several factors, including the lack of resources available to the various medical practices and the intensity and complexity of treatment ventures, limit the implementation of the DCCT approach to treatment. The DCCT used an entire team of professionals including physicians, nurses, diabetes educators, nutritionists, and behaviorists. Many medical practices are not able to provide such comprehensive services. Furthermore, the costs of these endeavors may be prohibitive.

It is evident that the described pharmaceutical care model provides a significant reduction in glycemia in the patient population studied. This model is applicable with site-specific modifications to a wide variety of practice settings, including community pharmacy, as it requires few resources to operate. However, some limitations of the study should be taken into consideration. First, sustained, long-term effectiveness of this model was not tested. The long-term advantages of this model remain to be evaluated. Second, the study indirectly compared our model with the standards of care provided by the internal medicine outpatient clinic at our institution. Standards of medical care may vary considerably among institutions. Third, the study required significant efforts by the investigators in terms of time allocations that may preclude management of a large patient load. Fourth, cost analysis of such an endeavor was not performed. Documentation of cost-effectiveness on a long-term basis by preventing costly diabetes-related complications would enhance the validity of such a model. Fifth, several barriers, which have been previously identified,^{16,23} may hinder the successful implementation of such a model. There will be a need for appropriate and ongoing training of pharmacists in providing comprehensive diabetes management that is consistent with the standards of care. Healthcare services have been primarily a physician territory. As a result of this historical role, physicians may view pharmaceutical care as an expansion into their territory and may be reluctant to delegate responsibility to other healthcare providers. Resources such as space and technical support are not readily available for pharmacist use in most clinical settings. Appropriate reimbursement guidelines do not exist. These issues or limitations need to be addressed for the development and implementation of pharmaceutical care.

The benefits of pharmacist interventions on outcomes of several disease states have been previously documented. McKenney et al.²⁴ demonstrated the effectiveness of pharmacist-provided clinical services in lowering blood pressure, enhancing compliance, and increasing patient knowledge in patients with hypertension. A literature review revealed a limited number of published studies on the impact of pharmaceutical care on diabetes management.¹¹⁻¹⁵ These studies have primarily focused on the impact of diabetes education, compliance enhancement, and drug therapy monitoring on disease outcomes. Although disease-specific outcome measures were not consistently obtained, these studies demonstrated that pharmacist-oriented services produced a favorable impact. Huff et al.¹¹ described a reimbursed pharmacist-managed educational service successfully provided over a 10-year period. Objective assessment of the impact of these services on disease outcome was not published. Schilling¹² described a pharmacist-run diabetes program where patients were seen by the pharmacist instead of physicians for routine diabetes management including education, compliance assessment, alteration of medication doses based on blood and urine glucose readings, and physician referral when appropriate. Although the report described the program in specific details, no analyses of glycemic control were provided. Hawkins et al.¹³ reported a randomized, controlled trial assessing the effectiveness of clinical pharmacist interventions on the management of patients with hypertension and diabetes in a pharmacist-run clinic setting. Although the pharmacist-rendered interventions were not specifically described, the

study results indicated that the experimental group experienced higher patient satisfaction attributed to higher kept-clinic-appointment and lower drop-out rates. Clinical outcome endpoint measures were not markedly altered by the pharmacist interventions. Sczupak and Conrad¹⁴ described a randomized, controlled study where the patients in the intervention group received expanded pharmacy services including therapeutic drug monitoring, education, and compliance training sessions provided at the time of regular physician appointments. The authors noted improved compliance rates and lower rates of hospital admissions as well as improved disease control evidenced by subjective reduction of symptoms of hyperglycemia, decreased postprandial blood glucose concentrations, and reduced urine glucose measurements, but no differences in fasting blood glucose concentrations. An abstract by Van Veldhuizen and Popovich¹⁵ compared a control group with an intervention group receiving either supplemental small group or individualized education sessions together with a 2-month follow-up period with a pharmacist. The educational efforts resulted in a marked increase in patient understanding of diabetes and its treatment, as well as better glucose control demonstrated by markedly reduced symptoms of hyperglycemia.

A number of associated metabolic disorders such as blood pressure elevation, lipid abnormalities, and renal disease often accompany diabetes. Although the correction of these disorders was not specifically targeted by this study, the emphasis placed on diet, exercise, and medication compliance along with the observed reduction in glycemia were expected to lead to improvement in these disorders in the intervention group. The lack of a significant change in these disorders, specifically the renal parameters, is most likely the result of the relatively short duration of the study.

The improvement in hyperglycemia and the enhanced patient understanding and engagement in diabetes control observed in our intervention group was not accompanied with any significant change in HRQOL assessment. Similar observations were previously reported where there was a lack of association between glycosylated hemoglobin measurements and HRQOL domains either on cross-sectional or longitudinal analyses.¹⁹ Improvement in health perceptions as a result of disease control may be offset by the tasks involving life-style changes undertaken by the patients in order to achieve tight glycemic control.

The incidence of reported mild-to-moderate hypoglycemia was approximately nine times higher in the intervention group compared with that in the control group. The higher risk is partly inherent to the study design and execution. Patients in the intervention group were repeatedly instructed and trained on the recognition and documentation of hypoglycemia. They were also questioned about the occurrence of these reactions at every clinic visit. Subjects in the control group were not given any specific instructions regarding hypoglycemia and data on its occurrence were collected at the end of the study where capture of such information may have been hindered by the duration of elapsed time.

Summary

This study establishes the effectiveness of pharmacist-provided pharmacotherapeutic interventions in the specific management of NIDDM. Patient-specific pharmaceutical

care should be considered an important component of medical care as its integration will improve patient outcomes. The impact of pharmaceutical care on the total healthcare costs of diabetes remains to be determined. $\simeq$

The study investigators thank Stewart Robertson MD for his time, effort, and support. The authors acknowledge George Grunberger MD for his valuable evaluation of this manuscript. We also thank the supportive staff of 4B general internal medicine clinic of Detroit Receiving Hospital–University Health Center.

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EXTRACTO

OBJETIVO: Evaluar la efectividad de un modelo de atención farmacéutica para el manejo de pacientes de origen afroamericano con diabetes no insulino dependiente (NIDDM), que asistían a un centro de atención ambulatoria. En este centro los pacientes eran atendidos por médicos internistas.

DISEÑO: Los pacientes elegibles para participar en el estudio se asignaron al grupo de intervención o al grupo de control utilizando un diseño aleatorio paralelo. Se les proveyó seguimiento por un período aproximado de 4 meses. Se excluyeron los pacientes no insulino dependientes que requerían el uso de insulina, los pacientes con disfunción renal, daño hepático o complicaciones cardíacas significativas, y los pacientes con incompetencia mental o historial de no cumplir con sus citas regulares al centro. Todos los aspectos relacionados al manejo de la diabetes los llevó a cabo un farmacéutico. Este modelo de cuidado incluía la evaluación de la farmacoterapia y los ajustes en dosificación, la educación comprensiva sobre diabetes y sus complicaciones, el adiestramiento para reconocer y tratar episodios de hipoglucemia e hiperglucemia, orientación sobre los medicamentos, instrucciones específicas sobre la dieta y el plan de ejercicio y adiestramiento para la automedición de su glucosa en sangre. Al inicio los pacientes se evaluaron semanalmente y luego cada 2-4 semanas. Los pacientes asignados al grupo de control continuaron en cuidado médico convencional. Sus citas médicas fueron cada 3-4 meses.

ESCENARIO: Un centro de atención ambulatoria afiliado a una institución educativa.

PARTICIPANTES: La población estudiada fue de pacientes afroamericanos residentes en áreas urbanas con NIDDM que asistían regularmente a un centro de atención ambulatoria.

MEDICIÓN DE RESULTADOS: Los resultados principales que se evaluaron fueron las concentraciones de glucosa en plasma en ayuna (FPG) y la hemoglobina glucosilada (GHb). Los resultados secundarios que se estudiaron incluyeron la presión sanguínea, creatinina sérica, depuración de creatinina, razón de microalbúmina/creatinina, colesterol total, triglicéridos, lipoproteínas de alta densidad y lipoproteína de baja densidad. Además se utilizaron medidas de calidad de vida para ambos grupos, al inicio y al finalizar el estudio.

RESULTADOS: Treinta y nueve pacientes finalizaron el estudio; 17 en el grupo de intervención y 22 en el control. El grupo de estudio consistió de 12 mujeres y 5 hombres con una edad promedio de 59 ± 12 años, peso total (TBW) de 93 ± 22 kg, índice de masa corporal (BMI) de 34 ± 7 kg/m² y duración de la diabetes de 6.8 ± 6.5 años. El grupo de control consistió de 15 mujeres y 7 hombres con edad de 65 ± 12 años, TBW de 88 ± 19 kg, BMI de 33 ± 7 kg/m² y duración de diabetes de 6.2 ± 4.8 años. Durante el período de estudio se llevaron a cabo 38 intervenciones farmacoterapéuticas en el grupo de intervención en relación con el régimen del hipoglucémico oral. En el grupo de control se informaron 9 intervenciones farmacoterapéuticas. El grupo de intervención mejoró en forma significativa. La hemoglobina glucosilada (GHb) al inicio fue de $11.5 \pm 2.9\%$ y el final de $9.2 \pm 2.1\%$ ($p = 0.003$), las concentraciones de

glucosa en plasma en ayuna mejoraron de 200 ± 72 mg/dL al inicio a 153 ± 42 mg/dL al final ($p = 0.015$). No hubo cambio en el grupo de control: GHb $12.2 \pm 3.5\%$ y $12.1 \pm 3.7\%$ ($p = 0.928$) y FPG 228 ± 85 mg/dL y 198 ± 71 mg/dL ($p = 0.054$), al inicio y al final respectivamente. Se obtuvieron diferencias significativas en la GHb final ($p = 0.003$) y FPG final ($p = 0.022$) en el grupo de intervención. No hubo cambios significativos en el control de presión sanguínea, perfil de lípidos, parámetros de función renal, peso o en las medidas de calidad de vida entre o dentro de los grupos.

CONCLUSIONES: Los resultados demuestran la efectividad del modelo de atención farmacéutica en reducir la hiperglucemia asociada a NIDDM en un grupo de pacientes afroamericanos residentes en áreas urbanas.

MIRZA MARTÍNEZ

RÉSUMÉ

OBJECTIF: Déterminer l'efficacité d'un programme de soins pharmaceutiques chez des afro-américains avec le diabète non-insulino-dépendant.

DEVIS EXPERIMENTAL: Les patients éligibles ont été randomisés dans un groupe témoin ou dans un groupe intervention pharmaceutique. Le suivi a été d'environ 4 mois. Le groupe intervention a reçu de l'enseignement sur le diabète, des conseils sur la médication, des renseignements sur la diète, l'exercice, et la surveillance du glucose sanguin à la maison ainsi que l'évaluation et l'ajustement des régimes hypoglycémiques actuels. Les médecins ont continué d'assumer le suivi médical usuel des patients du groupe témoin.

PARTICIPANTS: La population à l'étude consiste en un groupe d'afro-américains avec le diabète non-insulino-dépendant qui fréquente une clinique externe universitaire de médecine interne.

MESURES DE L'EFFET: Les principales mesures consistaient en une glycémie à jeûn et une hémoglobine glycosylée. Les mesures secondaires incluaient celles de la tension artérielle, la créatinine sérique, la clairance de la créatinine, le ratio microalbumine créatinine, et les valeurs du cholestérol total, des triglycérides, des lipoprotéins de haute densité et de faible densité. On a également mesuré la qualité de vie des sujets au début et à la fin de l'étude.

RESULTATS: Trente-neuf patients (17 du groupe intervention, 22 du groupe témoin) ont terminé l'étude. Douze femmes et 5 hommes formaient le groupe intervention. La moyenne d'âge $\pm$ la déviation standard était de 59 ± 12 ans, le poids corporel total (PCT) de 93 ± 22 kg, l'index de masse corporelle (IMC) de 34 ± 7 kg/m², et la durée du diabète non-insulino-dépendant de 6.8 ± 6.5 ans. Le groupe témoin comptait 15 femmes et 7 hommes avec un âge moyen de 65 ± 12 ans, un PCT de 88 ± 19 kg, un IMC de 33 ± 7 kg/m², et une durée du diabète de 6.2 ± 4.8 ans. Les valeurs de la glycémie se sont améliorées de façon très significative dans le groupe témoin passant pour la glycémie à jeûn ($p = 0.015$) et pour l'hémoglobine glycosylée ($p = 0.003$). Dans le groupe témoin, les valeurs de la glycémie n'ont point changé. Quant au contrôle de la tension artérielle, du profil lipidique, des paramètres de la fonction rénale, du poids, ou des mesures de la qualité de vie, on n'a noté aucune modification significative à l'intérieur des groupes ou entre ceux-ci.

CONCLUSIONS: Cette étude démontre l'efficacité des soins pharmaceutiques dans la diminution de l'hyperglycémie associée au diabète non-insulino-dépendant chez un groupe d'afro-américains.

LOUISE GAGNON